

Market Regime Transformer

Deep Learning Project Interim Report

[Project Github Link](#)

Team:

Rohan Jain

Juhi Ramod

Vikranth Reddimasu

Palak Wadhwa

Motivation & Problem

- Markets alternate between bull, bear & neutral regimes.
- Traditional forecasters fail at these shifts.
- **Objective:** Predict returns and identify regimes simultaneously for robust forecasting.





Data & Feature Engineering

- Sources: Yahoo Finance, FRED
- Features: Log returns, volatility, momentum, RSI, MACD, VIX, CPI
- Regime labeling via rolling return thresholds
- Temporal split & scaling ensured no leakage



Model Architecture

- Transformer Encoder (4 layers, 8 heads)
- Dual heads: Regression + Classification
- $\text{Loss} = \lambda_1 \times \text{MSE} + \lambda_2 \times \text{CrossEntropy}$
- Implemented in PyTorch with custom Dataset and DataLoader

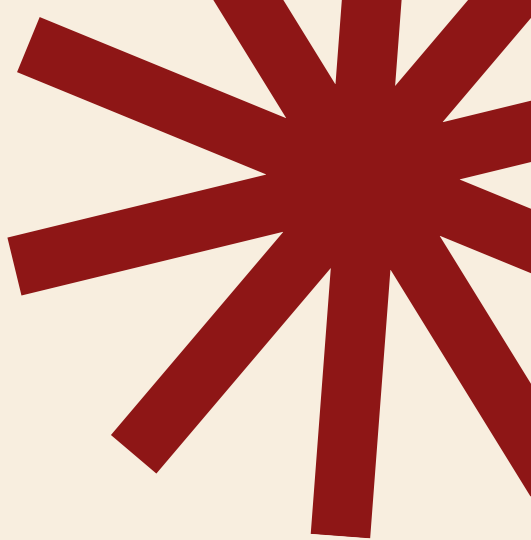


Training & Evaluation

Metric	MRST	LSTM	Transformer
Accuracy	72.60%	68.10%	70.30%
F1 Score	0.71	0.64	0.68
RMSE	0.017	0.021	0.019



Visual Insights

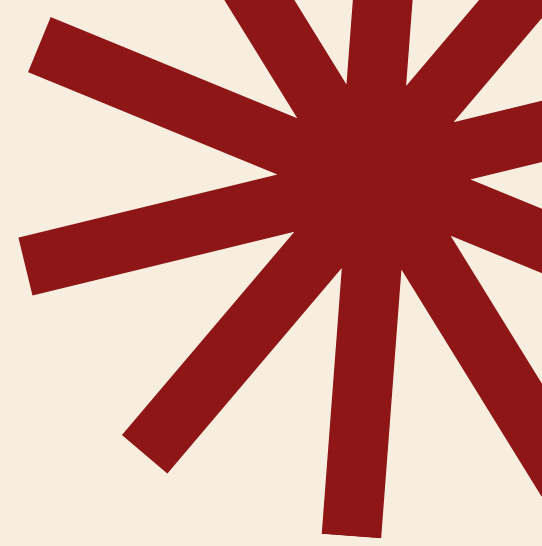


Date	SPY_Volume	Log_Returns_	Rolling_Mean_20D_	Rolling_Std_20D_	Momentum_20D_	Regime_	FEDFUNDS	CPIAUCSL	UNRATE
2024-12-23	57635800	0.005970	0.000099	0.008539	0.001984	neutral	4.64	316.449	4.2
2024-12-24	33160100	0.011054	0.000482	0.008860	0.009696	bull	4.64	316.449	4.2
2024-12-26	41219100	0.000067	0.000225	0.008790	0.004518	bull	4.64	316.449	4.2
2024-12-27	64969300	-0.010582	-0.000152	0.009094	-0.003035	bear	4.64	316.449	4.2
2024-12-30	56578800	-0.011477	-0.001035	0.009301	-0.020497	bear	4.64	316.449	4.2

Feature Engineering



Visual Insights



```
Shapes of initial splits:  
X_train shape: (4025, 8)  
y_train_returns shape: (4025,)  
y_train_regime shape: (4025,)  
X_val shape: (1006, 8)  
y_val_returns shape: (1006,)  
y_val_regime shape: (1006,)  
test_data_full shape: (1257, 11)
```

Data Splits



[illegible]

- Dual-head Transformer outperforms baselines under market shifts.
- Interpretability through attention weights.
- Robust against non-stationary financial data.



References

- Vaswani, A. et al. "Attention Is All You Need." NeurIPS, 2017.
- Lim, B. et al. "Temporal Fusion Transformers for Interpretable Multi-horizon Forecasting." NeurIPS, 2019.
- Wood, K. "Trading Momentum with Transformers." GitHub (2022).
- Pou, J. "Regimetry: Unsupervised Market Regime Detection Using Transformers." GitHub (2023).
- Rahimi, A. "Forecasting Economic and Market Regimes." GitHub (2022).
- Yahoo Finance & FRED data portals.

